

Host Intelligence for Airbnb Listings

Pricing, Demand, and Location-Based Recommendations

Final Project Report

1. Project Overview

This project analyzes Airbnb listing data across multiple U.S. cities to understand which listing features are connected to stronger demand.

The main goal is to help answer a practical host and marketplace question: *What type of Airbnb listing is most likely to perform well in a specific location?*

The project looks at pricing, location, property size, room type, amenities, host behavior, and product-style features. It also builds a location-based recommendation workflow that can compare nearby listings and recommend stronger listing types for a selected latitude, longitude, and radius.

The final output is not just a model. It's a first version of a host intelligence tool. A user can enter a location, and the workflow returns top-performing local listing types, pricing patterns, amenity recommendations, ADR-supporting amenities, and predicted monthly demand.

Room Type	Bedrooms	Listings	Avg Reviews/Month	Predicted Reviews/Month	Avg Price	ADR-Supporting Amenities
Entire home/apt	0.0	66	1.57	1.42	161.71	[private entrance, fan, air conditioning, bed ...
Entire home/apt	1.0	134	1.45	1.13	188.99	[toiletries, kids amenities, air conditioning,...
Entire home/apt	4.0	35	1.32	0.42	718.63	[beach access, bbq grill, patio balcony, hot t...

2. Business Problem

Airbnb hosts need to understand how to position their listings in a specific market. A property that works well in one location might not work well in another. For example, a downtown studio might perform well because of price, walkability, and convenience. A larger family-friendly property might perform better near theme parks, beaches, or suburban attractions.

From Airbnb's perspective, better host guidance could improve marketplace quality. **If hosts understand what guests are looking for in a given area, they can price more competitively, add better amenities, improve their listings, and attract the right type of guest.**

This project is focused on three stakeholder questions:

- For hosts: What listing attributes are associated with stronger demand?
- For guests: Which listings are easier to compare, trust, and book?

- For Airbnb: How could listing-level data support better host recommendations and marketplace performance?

3. Data Source and Target Variable

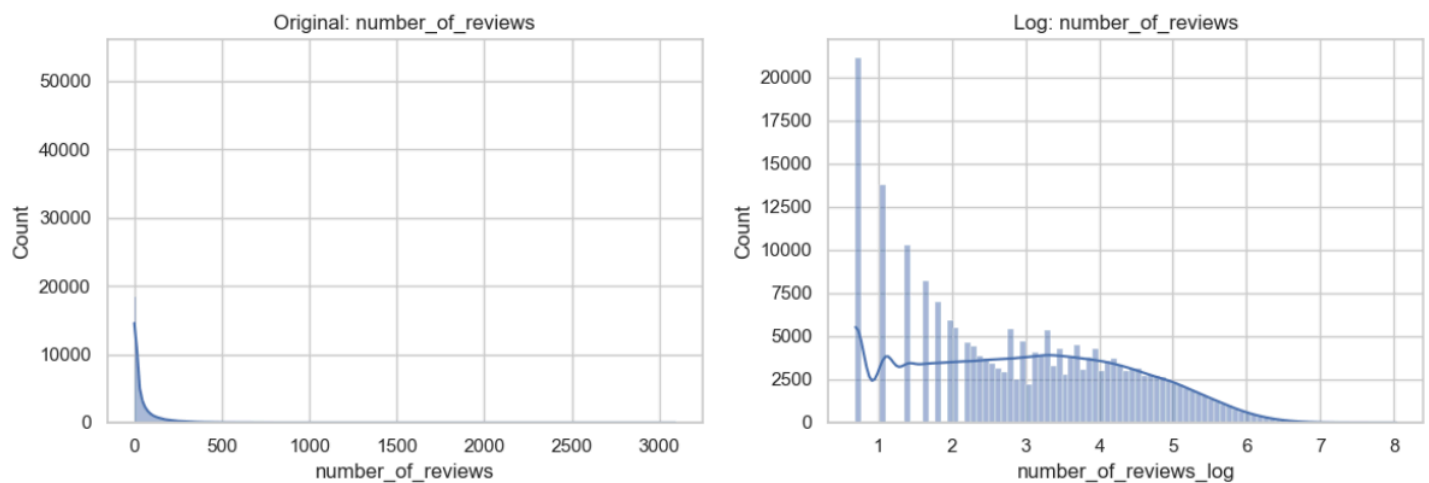
The data comes from Inside Airbnb via Kaggle. The dataset includes listing-level information such as price, location, room type, bedrooms, bathrooms, amenities, host details, reviews, and review activity.

The original target variable was **number_of_reviews**, however, after inspection I realized that it was unreliable since some listings had been active for much longer. The only metric I had with some datetime information was monthly reviews. From there, I updated the main target variable to be **reviews_per_month**.

Some issues I ran into: A review doesn't equal a booking, some bookings receive multiple reviews, and not every guest leaves a review. However, it gives a useful signal for comparing active listings.

A listing with more reviews is likely receiving more guest activity than a similar listing with fewer reviews per month.

Originally I used `np.log1p(number_of_reviews)` because the original target is skewed. After prediction, the results are converted back to normal reviews per month using `np.expm1()`. Eventually I capped the reviews to `<50 / month` because I still wanted to interpret the demand signal, but didn't want heavily skewed data.



4. Data Cleaning and Wrangling Decisions

The first major cleaning step was combining raw Airbnb listing files across multiple cities and creating a consistent dataset.

Important cleaning decisions included:

- Standardizing column names and data types

- Cleaning price values into numeric format
- Cleaning bathroom, bedroom, bed, and accommodates fields
- Standardizing amenity strings into cleaner amenity categories
- Creating binary amenity columns
- Handling missing values
- Removing listings with missing or unusable demand signals for modeling
- Creating separate datasets for broad market context and modeling

One important issue was reviews_per_month. In the full dataset, missing reviews_per_month values had been filled with 0. This was useful for broad market context, but not for demand modeling. A missing value doesn't mean the listing had zero demand. It could mean the listing had no review history or was not active long enough to show demand.

To fix this, I created two main datasets to make the modeling target more reliable:

- **airbnb_full.csv:** This includes the full dataset and fills missing reviews_per_month values with 0. It's helpful for an overall view.
- **airbnb_clean.csv:** This removes listings where reviews_per_month is missing or 0. This dataset is used for EDA and modeling because it only includes listings with a usable demand signal.

```
Full dataset shape: (268387, 75)
Clean demand dataset shape: (212027, 75)
```

Original Amenity Data

```
["Coffee maker", "Cleaning products", "Extra p...
["Shampoo", "Hair dryer", "Coffee maker", "Pet...
["Fire extinguisher", "Refrigerator", "Microwa...
["Shampoo", "Electric stove", "Hair dryer", "C...
["Shampoo", "Hair dryer", "Coffee maker", "Cle...
```

amenity

```
65" hdtv with standard cable, amazon prime vid...
```

```
fast wifi – 916 mbps
```

```
40" hdtv with fire tv, hbo max, netflix
```

```
hdtv with amazon prime video, disney+, fire tv...
```

```
dove/ suave body soap
```

Cleaned Amenity Data

amenity_wifi	263384
amenity_smoke_alarm	252143
amenity_kitchen	239734
amenity_tv	237772
amenity_essentials	234797
amenity_air_conditioning	224928
amenity_hangers	212656
amenity_heating	210827
amenity_hair_dryer	209333
amenity_carbon_monoxide_alarm	209193
amenity_refrigerator	205397
amenity_cooking_basics	204977
amenity_hot_water	202033
amenity_toiletries	201896
amenity_iron	201649
amenity_parking_free	201387

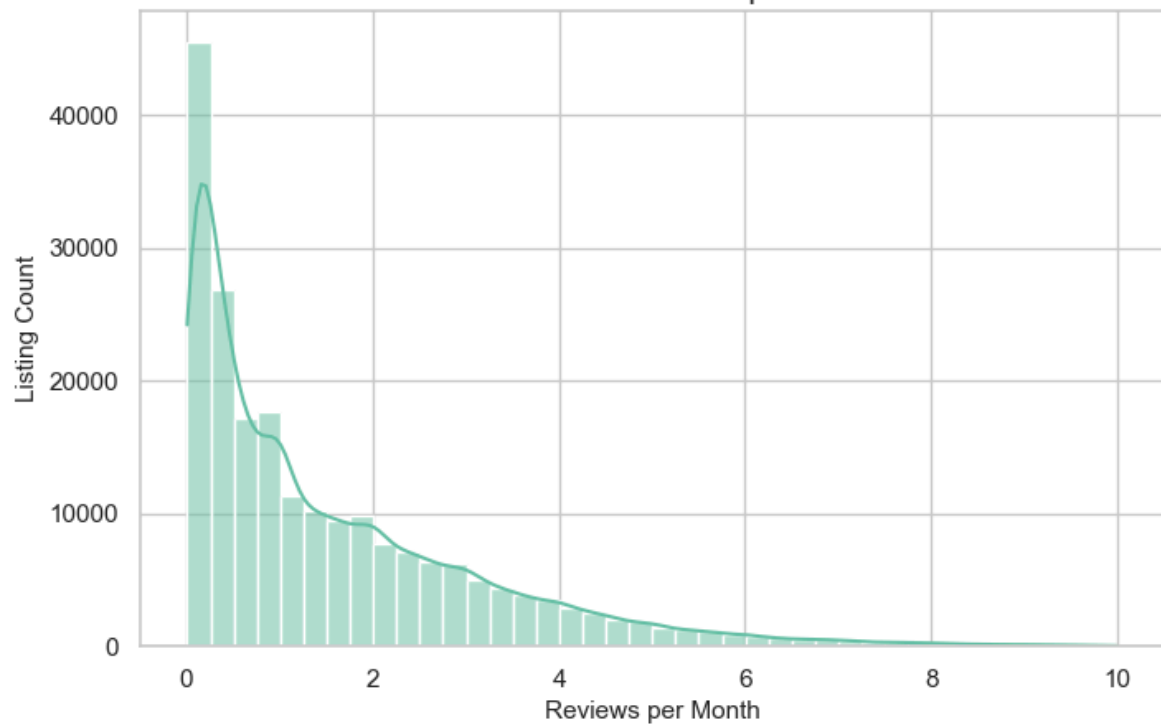
5. Exploratory Data Analysis

The EDA showed that Airbnb demand is uneven and location-sensitive. Some listings have very high review activity, while many have lower monthly demand. This makes demand harder to predict with a simple linear model.

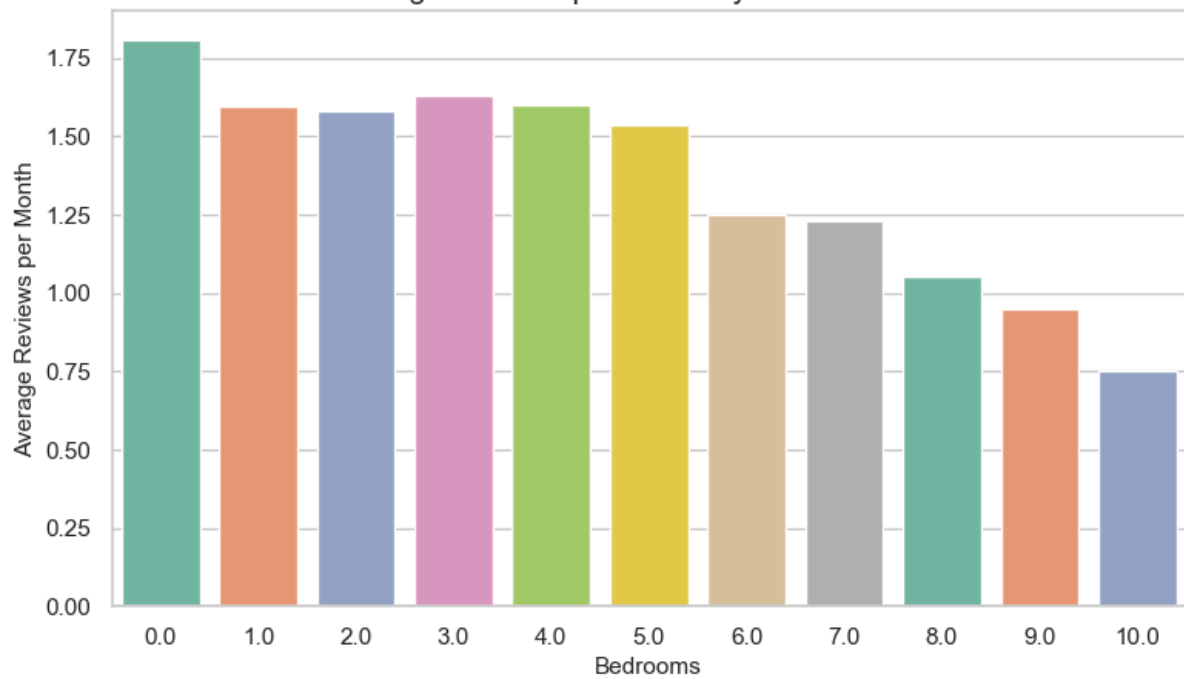
Patterns that stood out:

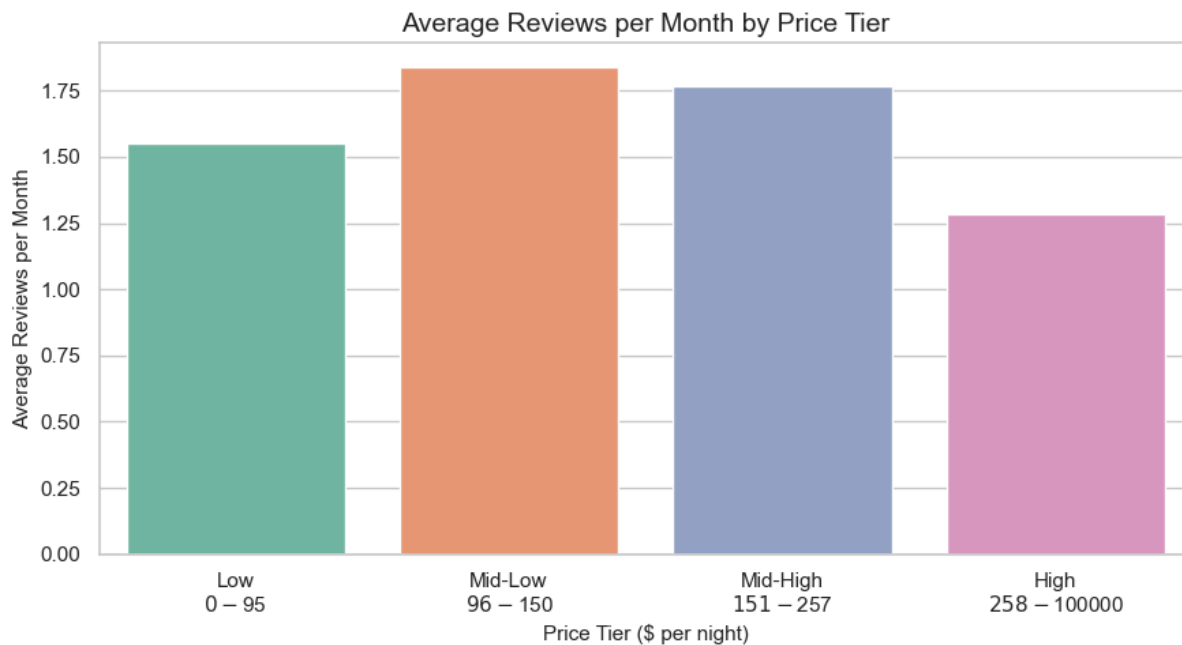
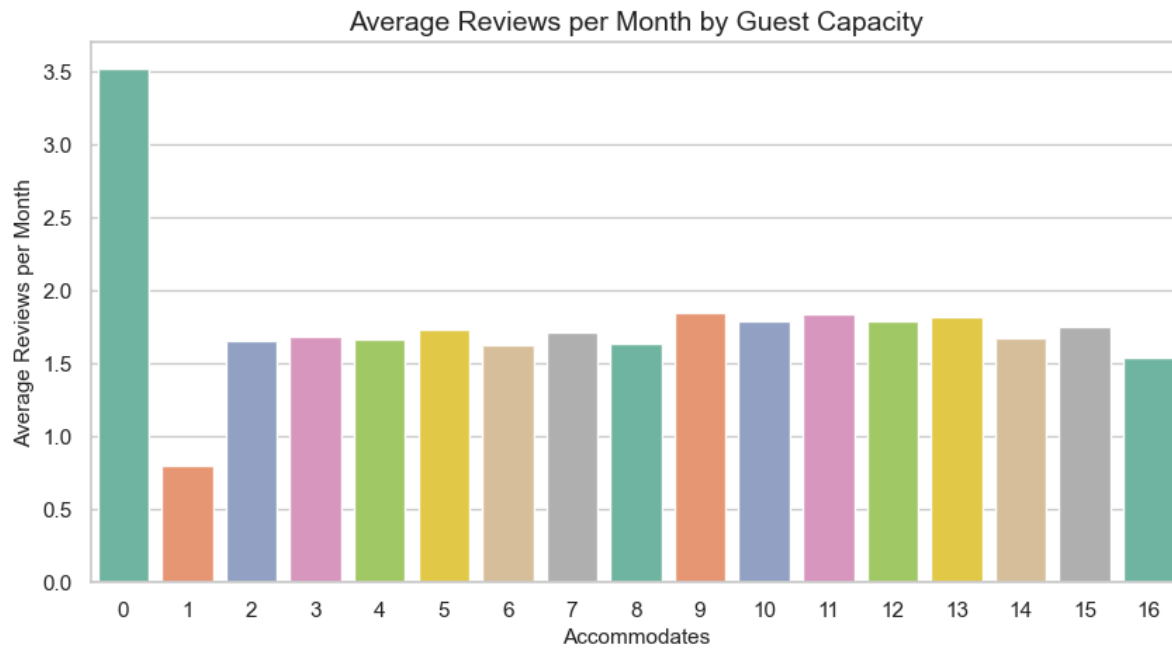
- **Room type:** Entire homes and apartments often perform differently from private rooms, shared rooms, and hotel rooms.
- **Bedroom count and guest-capacity:** In some local markets, studios or smaller units perform well because they are affordable and convenient. In other markets, larger homes perform better because they support families or groups.
- **Pricing:** Listings that are too expensive compared with local comps often have weaker demand unless they offer stronger amenities, better location, or a more specific guest fit.
- **Amenities:** Some amenities are standard expectations, like WiFi, bed linens, coffee maker, and hot water. Other amenities contribute to a higher ADR, like free parking, a full kitchen (vs kitchenette), patio/balcony, city views, or hot tub, depending on the market.
- **Location:** Demand patterns shift by city and neighborhood, so a recommendation tool needs local comp logic instead of one broad rule for all markets.

Distribution of Reviews per Month



Average Reviews per Month by Bedroom Count





6. Feature Engineering

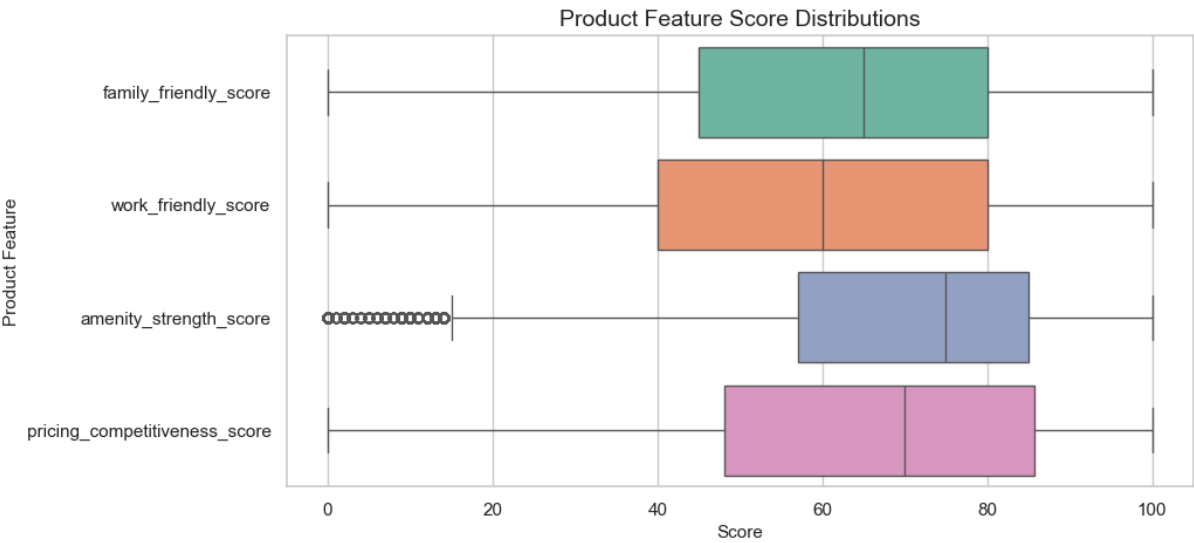
A major part of this project was turning raw listing data into more useful product-style features.

Instead of only using raw amenities and host fields, I created features that could support host recommendations and marketplace insights.

I also created dummy and binary features for categorical and amenity-based variables where needed. This allowed the models to use listing categories, host signals, and amenity signals in numeric form.

Feature	Description
price_clean	Cleaned nightly price used for analysis and modeling.
pricing_competitiveness_score	A score that compares a listing's price against similar listings. Higher scores mean the listing appears more competitively priced.
family_friendly_score	A score based on family-related amenities and signals. Higher scores mean the listing seems more set up for families.
work_friendly_score	A score based on work-friendly signals like WiFi, workspace, and longer-stay support.
amenity_strength_score	An overall score that captures how well-equipped a listing is.
host_response_time_ord	An ordinal version of host response time. Higher values represent faster or stronger response behavior.
host_is_superhost	A binary feature showing whether the host is a Superhost.
instant_bookable	A binary feature showing whether the listing can be booked instantly.

These features make the project more useful because they translate raw listing data into signals that a host or product team could understand.



Feature	Mean	Median	Std Dev	Min	Max
family_friendly_score	60.74	65.0	25.88	0.0	100.0
work_friendly_score	60.05	60.0	26.52	0.0	100.0
amenity_strength_score	70.81	75.0	21.66	0.0	100.0
pricing_competitiveness_score	62.58	70.0	29.46	0.0	100.0

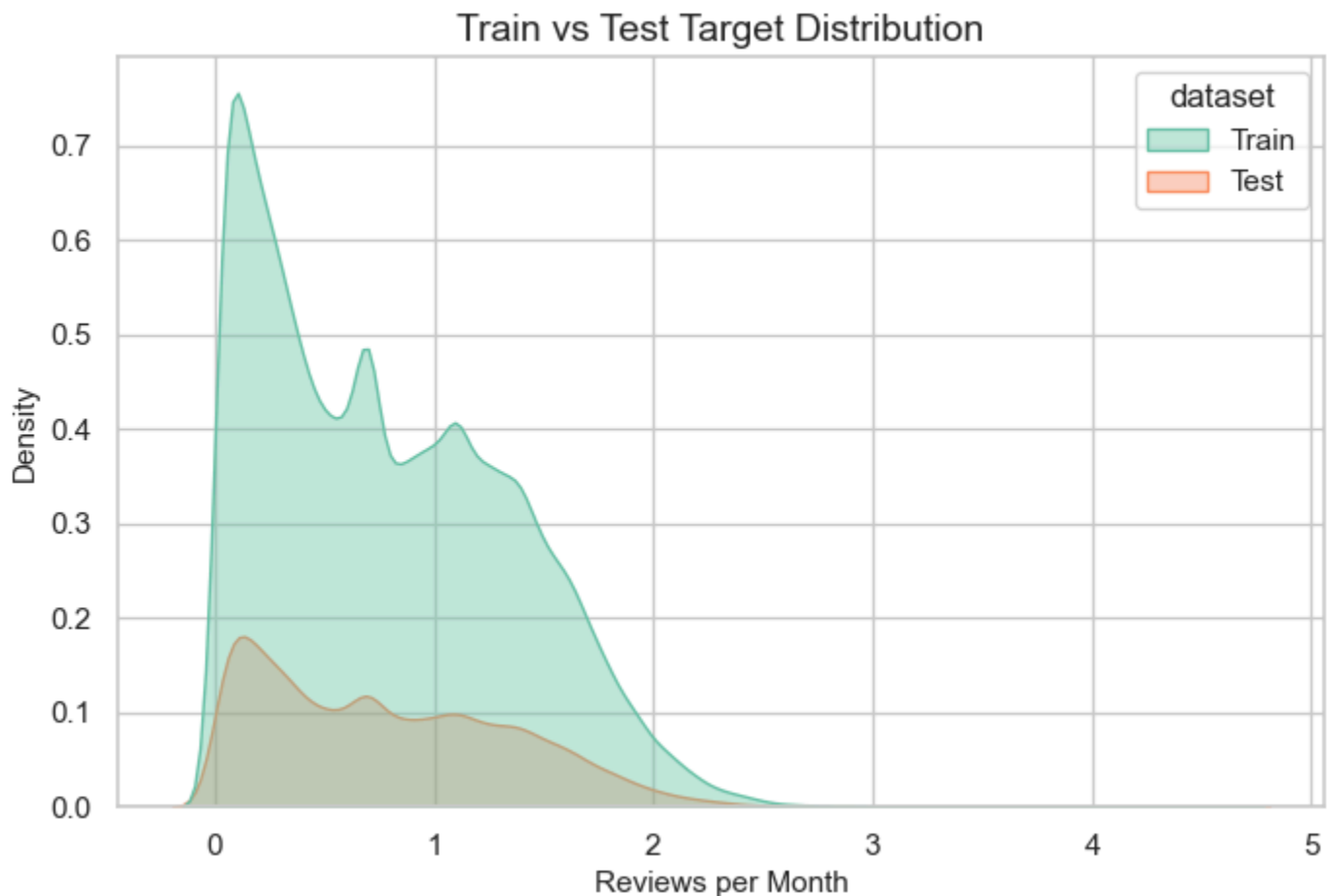
7. Preprocessing, Train/Test Split, and Scaling

Before modeling, the data was split into training and testing subsets so model performance could be checked on unseen data.

Numeric features were scaled where needed for linear and regularized models.

Tree-based models, like Random Forest, did not require scaling in the same way, but the preprocessing workflow kept the modeling process consistent.

The target variable was log-transformed using `np.log1p(reviews_per_month)`. This helped reduce skew and made the model less dominated by extreme high-demand listings.



8. Modeling Approach

The modeling goal was to predict monthly demand using `reviews_per_month` as the target.

Several models were tested:

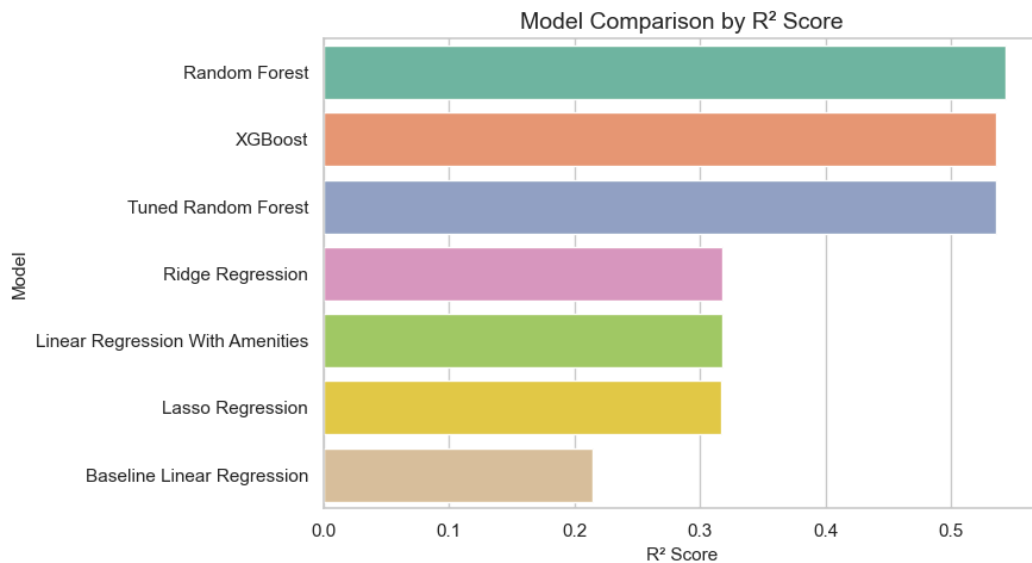
- Linear Regression
- Ridge Regression
- Lasso Regression
- Random Forest Regression

Linear Regression was useful as a baseline. Ridge and Lasso helped test whether regularization improved performance. Random Forest was tested because demand is likely shaped by nonlinear patterns across location, price, host signals, amenities, and property characteristics.

The final model selected was a **tuned Random Forest model**.

The final model uses a recommendation-focused feature set:

- `price_clean`
- `pricing_competitiveness_score`
- `accommodates`
- `bedrooms`
- `beds`
- `bathrooms`
- `latitude`
- `longitude`
- `host_is_superhost`
- `host_response_time_ord`
- `instant_bookable`
- `family_friendly_score`
- `work_friendly_score`
- `amenity_strength_score`



9. Model Results

The Random Forest model performed better than the simpler baseline models because it could pick up more complex patterns in the data.

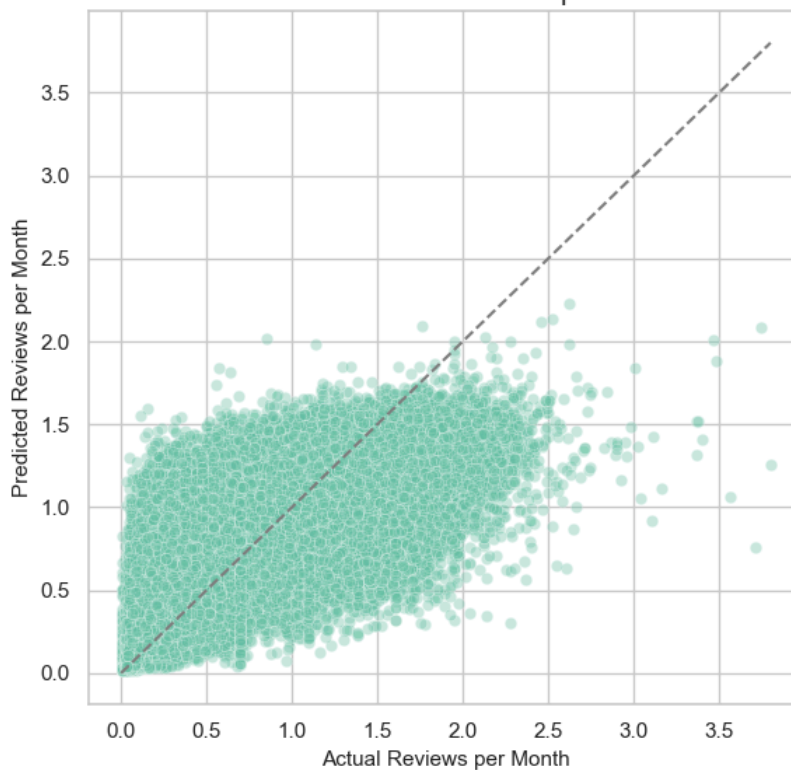
The final model was evaluated using regression metrics:

Metric	Final Value	Plain-language meaning
R²	0.536	How much variation in demand the model explains.
MAE	0.296	Average prediction error in reviews/month terms.
RMSE	0.387	Prediction error that penalizes larger misses more strongly.

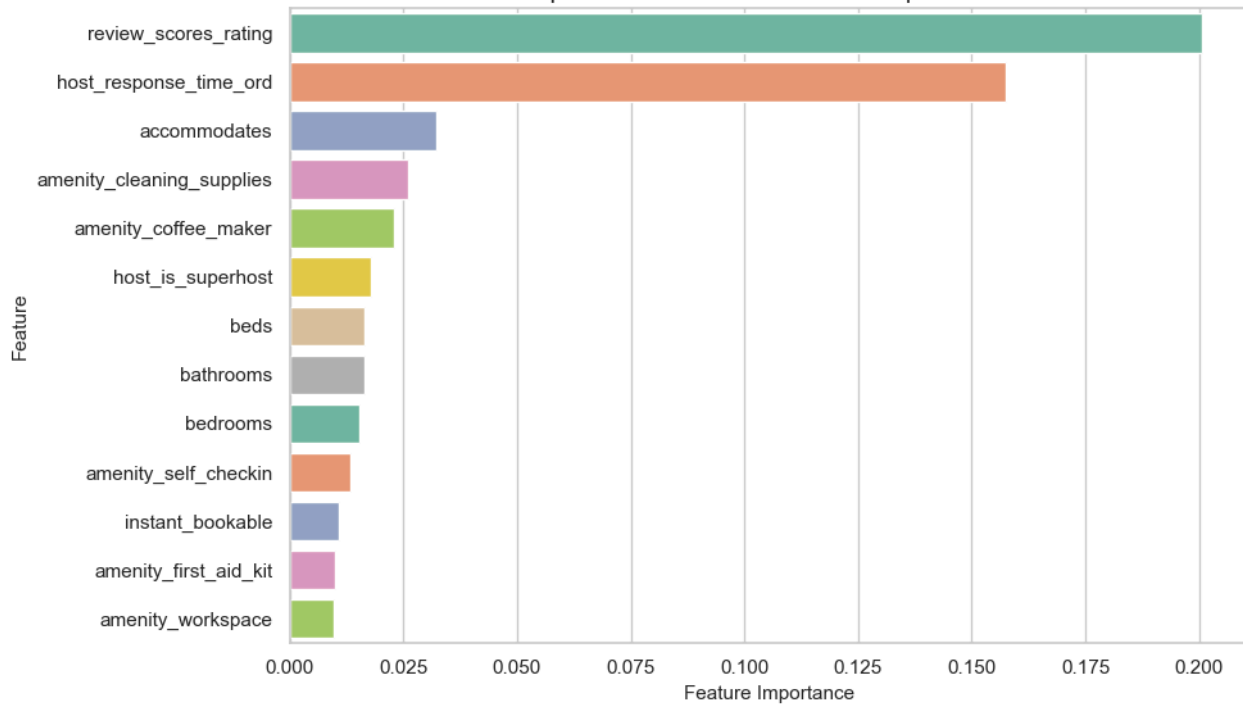
The final model metrics and model comparison table were also saved as a separate model metrics file for submission. The model gives a useful estimate of demand, but it is not perfect. One important limitation is that it tends to underpredict very high-demand listings. This means the model is better at estimating typical demand than extreme outliers.

The actual vs predicted plot showed that the model can capture broad demand patterns, but it doesn't perfectly match every listing. This is expected because Airbnb demand is affected by factors that are not fully captured in the dataset, such as listing age, seasonality, true bookings, photo quality, interior design quality, events, and neighborhood-specific demand.

Actual vs Predicted Reviews per Month



Top 15 Random Forest Feature Importances



10. Key Model Insights

The model and feature analysis showed that demand is shaped by a mix of location, price, listing size, host signals, and product quality.

- **Pricing competitiveness:** guests are likely comparing listings against similar local options.
- **Location:** the same listing type can perform differently depending on the market.
- **Host trust signals:** Superhost status, responsiveness, and instant booking can make a listing feel easier and safer to book.
- **Amenity strength:** Some amenities are baseline expectations, while others might help support higher ADR or stronger positioning.

The model can estimate which amenities are price drivers and at what rate:

	amenity	overall_rate	high_price_rate	rate_lift
29	kitchen	0.629	0.924	0.295
34	oven	0.345	0.606	0.261
45	stove	0.405	0.652	0.246
10	dishwasher	0.167	0.379	0.212
30	long term stays	0.299	0.424	0.125
35	parking free	0.598	0.712	0.114
37	patio balcony	0.417	0.530	0.114

11. Location Recommendation Prototype

The final notebook turns the analysis into a reusable location recommendation workflow.

The user enters:

- Location name
- Latitude
- Longitude
- Radius
- Number of top listing types to return

Workflow + Output:

1. Finds nearby Airbnb listings within the radius
2. Summarizes local comp performance
3. Identifies top-performing listing types

4. Compares pricing and demand
5. Finds standard amenities
6. Finds ADR-supporting amenities
7. Adds model-predicted reviews per month
8. Returns a final recommendation table

The recommendation output includes room type, bedroom count, listing count, average reviews per month, median reviews per month, average price, average accommodates, average bathrooms, standard amenities, ADR-supporting amenities, possible niche amenities, and predicted reviews per month.

	Room Type	Bedrooms	Listings	Avg Reviews/Month	Median Reviews/Month	Avg Price	Predicted Reviews/Month	ADR-Supporting Amenities
2	Private room	1.0	27	1.30	1.04	159.19	1.40	tv, air conditioning, host services, bbq grill
1	Entire home/apt	0.0	56	1.62	1.36	173.66	1.32	private entrance, air conditioning, bed linens...
0	Entire home/apt	1.0	100	1.70	1.32	217.54	1.17	air conditioning, workspace, toiletries, carbo...

12. Example Recommendation: Downtown San Diego

For downtown San Diego, the model and local comps pointed toward a budget studio as one of the strongest listing strategies.

Studios had strong demand, and lower to mid-low priced listings performed better than high-priced listings. The best strategy looks like a compact, well-priced studio with strong basics, easy check-in, work-friendly features, and a few ADR-supporting amenities like full kitchen features, parking, patio/balcony, city view, gym access, or hot tub.

	scenario	predicted_reviews_per_month	price_clean	pricing_competitiveness_score	accommodates	bedrooms	bathrooms	family_friendly_score	work_friendly_score
0	Budget studio	1.84	125	75	2	0	1	0	3
2	Mid-priced 2-bedroom	1.53	225	65	5	2	2	5	4
1	Standard 1-bedroom	1.45	160	70	3	1	1	2	4
3	Group-friendly 3-bedroom	1.21	325	60	8	3	2	6	4

13. Market Theme Maps

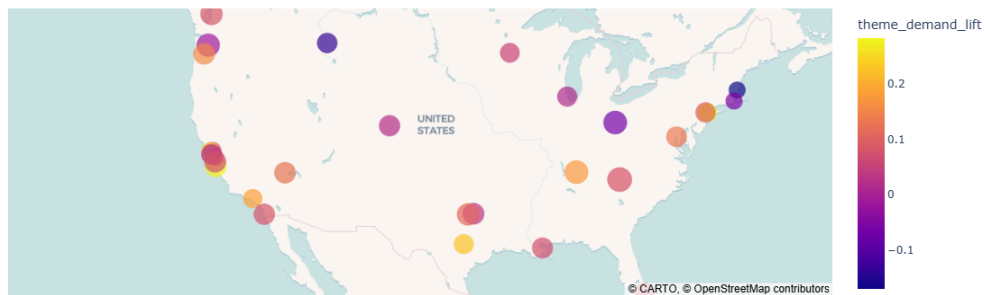
I also created broader market maps to compare city-level demand patterns for different amenity themes.

The themes included:

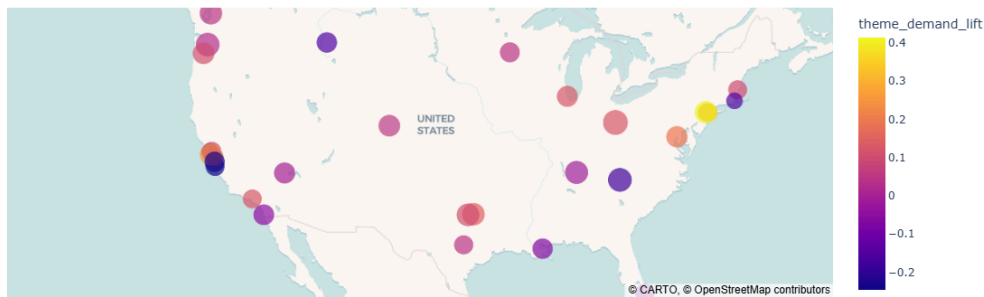
- **Family-friendly demand:** uses family-friendly score and guest capacity to identify listings that seem more family-oriented.
- **Nature/view demand:** uses amenities related to views, beach access, water, gardens, patios, balconies, and similar outdoor or nature signals.
- **Bachelorette/group-trip demand:** A listing had to accommodate at least 6 guests and include at least one social or experience-based amenity, such as hot tub, pool, patio, balcony, BBQ/grill, beach access, or sound system.

These maps are exploratory. They are useful for seeing broad patterns, but they should not be treated as final proof that a theme causes stronger demand.

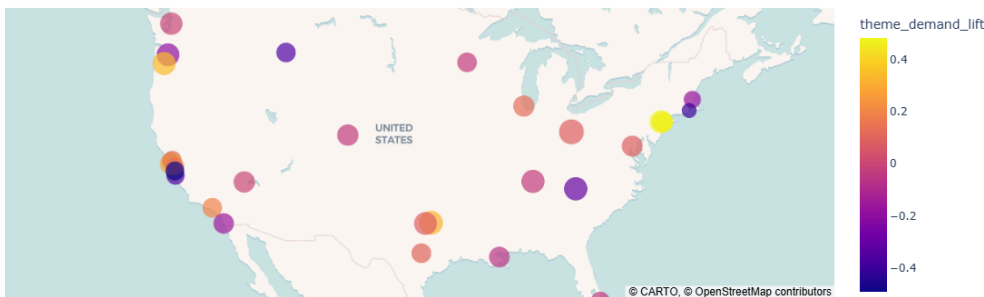
Top Cities for Nature & Views Demand



Top Cities for Family-Friendly Demand



Top Cities for Bachelorette Demand



14. Key Findings

- Airbnb demand varies by market.
- Listings that feel like a good value compared with local comps tend to have stronger demand.
- Smaller listings can perform very well in dense urban markets. For example, downtown San Diego showed strong demand for studios.
- Family-friendly features are not automatically a demand advantage. They need to match the location, guest type, and property size.
- Amenities should be separated into standard amenities and ADR-supporting amenities. Standard amenities help a listing compete. ADR-supporting amenities might help justify a higher nightly rate.
- The recommendation prototype is useful because it combines local comps with model predictions. Local comps show what is already working nearby, while the model adds a predicted demand estimate.

15. Limitations

- **reviews_per_month** is a placeholder for demand. It does not show true bookings, occupancy, revenue, or conversion rate. It can also be noisy because multiple guests from the same trip may leave reviews.
- Listing age is not fully accounted for. Older listings may have more reviews because they have been active longer, not necessarily because they perform better.
- The model works best in markets represented in the training data. If a location has few or no nearby comps, the recommendation is less reliable.
- The dataset does not fully capture seasonality or location context. The model does not directly know distance to beaches, theme parks, convention centers, transit, events, or major attractions.
- Some important listing quality signals are missing. The model does not fully capture photo quality, interior design, views, condition, listing copy, or actual guest trip purpose.

16. Future Improvements

- **Booking data:** Add actual bookings, occupancy, ADR, RevPAR, and revenue data so demand and performance are measured more directly.
- **Listing age:** Add listing age so the model can separate true demand from how long a listing has been active.
- **Seasonality:** Add summer demand, holiday demand, local events, and market-specific travel patterns.
- **Location context:** Add distance to beaches, theme parks, convention centers, parks, transit, nightlife, downtown areas, and major attractions.
- **Property input tool:** Build a version where a user can enter a property address or Zillow/Redfin listing, then get a suggested Airbnb positioning strategy.

- **Image analysis:** Use CV to analyze images for comparing demand between similar property types to see a correlation between image/interior design and demand.

17. Measurement Strategy

If Airbnb wanted to test a host recommendation tool, it could use an A/B test.

Control Group: Uses standard host dashboard

Variable Group: Hosts would receive personalized recommendations for pricing, amenities, listing copy, host trust signals, and visual presentation.

**Hosts would have to have similar metrics (host status) and listings*

Primary success metrics could include:

- Booking conversion
- Nights booked
- Revenue per available night
- Host adoption of recommendations

Secondary metrics could include:

- Review volume
- Guest satisfaction
- Host retention
- Listing update rate
- Search-to-book conversion

Guardrail metrics could include:

- Cancellation rate
- Guest complaints
- Host dissatisfaction
- Price volatility
- Decline in perceived listing quality

18. Final Business Recommendation

The main takeaway from this project is that Airbnb performance is not just about having a “nice” property or setting the right price. It is about understanding what kind of property makes sense for that specific location.

I wanted to build this tool because, as a potential investor, I would want to know what type of property to buy, how to position it, and what features would actually matter in that market. A studio downtown, a family-friendly home near attractions, and a group-friendly house near nightlife are all different strategies.

The location recommendation prototype is the strongest part of this project because it starts to answer that question directly. It uses nearby comps, pricing patterns, amenities, host signals, and model predictions to give a more practical recommendation.

A future version of this tool could help investors or hosts decide not just what to charge, but what kind of Airbnb product to create in the first place.

How the client can use these findings:

- 1. Host intelligence tool:** The tool could help hosts understand which listing type, pricing range, amenities, and trust signals fit their local market.
- 2. Listing simulation tool:** Let hosts or investors test different listing setups before launching. They could enter location, ADR, amenities, guest capacity, host settings, and photo quality, then see how the property might perform and how it should be positioned in that market.
- 3. Market theme strategy:** Group listings and markets by themes like family-friendly, work-friendly, nature/view, urban studio, or group-trip positioning. This makes the data easier to use for host guidance, investment decisions, and market-specific recommendations.